title

1 Hello everyone! Today, I will be presenting on characterizing Double White Dwarf star systems—specifically how we use multi-messenger astrophysics and electromagnetic observations to prepare for the upcoming LISA mission.

2

- Two degenerate stellar remnants orbiting a center of mass
- Ultra compact detached systems with short (few minutes) orbital periods create monochromatic gravitational radiation
  - Which produce ripples through space time
- WD lacks nuclear burning so their evolution is driven by gravitational wave angular momentum loss -> orbits get so short they can collide.

3

- LISA (Laser interferometer space antenna) GW detector in space
  - Address the issue of noise from earth tectonic motion
- Why does a laser in space need optical observations from telescopes?
  - Known DWD can act as verification binaries for LISA
  - Optical surveys give prior measurements
    - Sky positions, orbital periods and inclinations which can be used to predict the systems GW signal
  - Allowing us to validate performance and calibrate LISA in the first few months of operation

4

- Determine physical properties such as mass and tilt
  - Fit the optical light curves using photometric modeling software
- Systems fall into distinct geometric categories
- J0651 an eclipsing system:
  - Period of ~12 minutes, measured using the V-shape spacing in the plot which are the clear contact points for the secondary and primary source
  - directly constrains orbital tilt and size to precise percentage
  - GW forecast for these types: exceptionally strong
    - LISA will easily detect

4

- J0106 an ellipsoidal system:
  - Light variation caused by tidal stretching
    - The stars stretch each other to an oval like shape, similar to how the moon affects earths tides
  - No eclipsing:
    - Lose edge on timing constraints
    - Mass uncertainty of ~10 to 20 percent
    - Relies heavily on spectroscopic priors

5

- Lastly J2322 a featureless system:
  - Caused by huge brightness contrast
    - In optical and UV WL the hot primary (16,000 to 20,000 kelvin) drowns out the cooler secondary (7,000 to 10,000 kelvin)
  - Causing there to be no clear modulations leaving orbital tilt in the model completely unconstrained (~80% uncertainties)

5

  - Major problem with GW signal
    - because the wave strength we detect on Earth depends heavily on how the binary orbit is tilted relative to us
  - unknown tilt angle creates a mathematical ambiguity between the tilt and the system's mass
    - inflates error bars on the secondary mass to over 50 percent
  - Main issue with ~50% of targets being this type

6

- My research:
  - Address the issue of obscuring the secondary
    - Using infrared to observe secondary because it contributes a much larger fraction of light in the IR than in the optical or UV
  - I am using tools like PanSTARRS-1 to observe it the sources has any other stars appearing by it in the infrared
  - Then using Sphere-X to observe the light curve of the system in the infrared

6

- Thus my over all project is addressing that over half of optical DWD sources lack clear geometric modulations
  - So we extend our observations into the infrared to uncover the hidden secondary star
- Provides a missing physical constraint needed to break optical degeneracies and provide precise priors to use in LISA calibration.